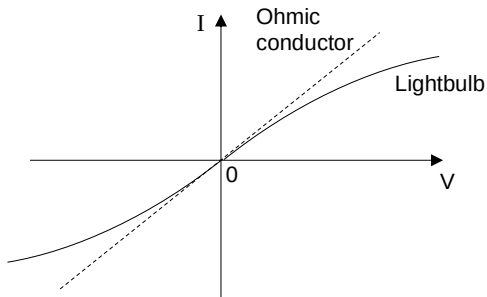


10(a)(i) e.m.f. is the amount of other forms of energy converted to electrical energy per unit charge delivered by the source of e.m.f, while p.d. is the amount of electrical energy converted to other forms of energy per unit charge flowing between 2 points in the circuit.

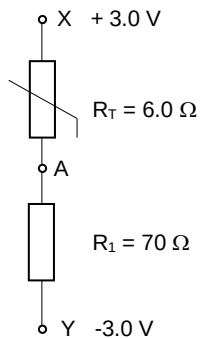
(ii)1. There are electrons throughout the entire circuit, including inside the lightbulb. So when the switch is closed, the lightbulb immediately experiences a flow of electrons.

21.



- II. The ohmic conductor has a constant resistance. Compared to the ohmic conductor, the lightbulb has a bigger V for the same I . Since $R = V/I$, the lightbulb's resistance increases with increasing I .

(b)(i)



Potential divider principle:

$$\frac{V_{XA}}{V_{XY}} = \frac{R_T}{R_T + R_1}$$

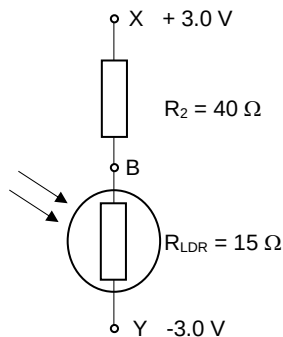
$$V_{XA} = V_{XY} \frac{R_T}{R_T + R_1} = 6.0 \times \frac{6.0}{6.0 + 70}$$

$$= 0.4737 \text{ V}$$

$$V_A = 3.0 - 0.4737 = 2.526$$

$$\approx 2.5 \text{ V}$$

(ii)



Potential divider principle:

$$\frac{V_{XB}}{V_{XY}} = \frac{R_2}{R_2 + R_{LDR}}$$

$$V_{XB} = V_{XY} \frac{R_2}{R_2 + R_{LDR}}$$

$$= 6.0 \times \frac{40}{40 + 15}$$

$$= 4.3636 \text{ V}$$

$$V_B = 3.0 - 4.3636 = -1.3636$$

$$\approx -1.4 \text{ V}$$

(iii) p.d. across lightbulb = $V_A - V_B$
= $2.526 - (-1.363)$
= $3.9 \text{ V} > 3.5 \text{ V}$

Also, since $V_A > V_B$, the diode is in forward bias, and can conduct.
So the lightbulb will light up.

(iv) For the lightbulb to light up, V_A should be as high as possible. So V_{XA} should be as small as possible. So using potential divider principle, R_T should be as small as possible.

So the temperature should be high.

V_B should be as low as possible. So V_{XB} should be as high as possible. By potential divider principle, R_{LDR} should be as low as possible.
So the brightness should be high.